General Chemistry I

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Lecture No. 7

Lewis structure

7. Lewis structure

- What is the molecular structure of NH_3?

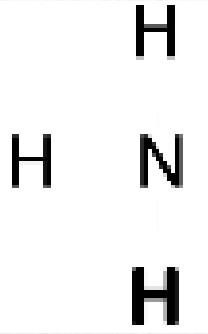
OR

How the N and H atoms are bonded together in NH_3 molecule?

7. Lewis structure

A Step-By-Step Approach To Writing Lewis Structures:

<u>Step 1:</u> Decide which atoms are bonded



<u>Step 2:</u> Determine the total number of valence electrons in a molecule

Valence electrons of N = 5 e^-

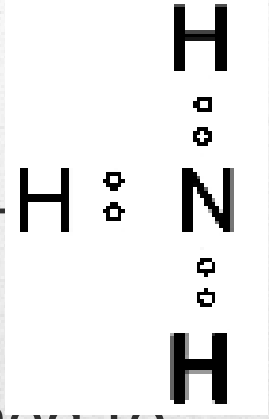
Valence electrons of H = 1 e^- x 3 atoms

Total number of valence electrons = 8

7. Lewis structure

A Step-By-Step Approach To Writing Lewis Structures:

<u>Step 3:</u> place two electrons between each two atoms (i. e. in each bond)



Note that: your placed 6 electrons out of 8 elec

<u>Step 4:</u> complete the octet of the atoms attached to the central atom by adding electrons in pairs.

Note that: no electrons will be placed in this step, where each bonded atom is hydrogen atom with 2 electrons.

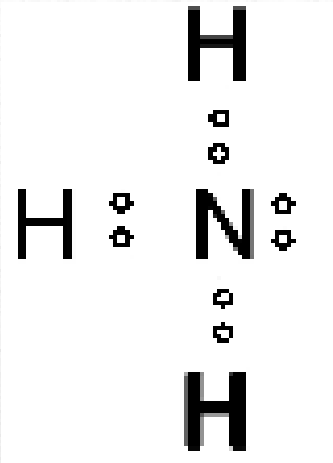
7. Lewis structure

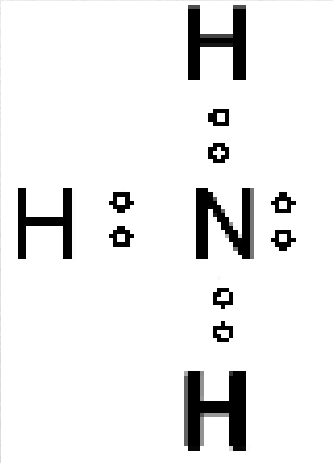
<u>Step 5:</u> Place any remaining electrons on the central atom in pairs.

Note: two electrons were placed around the central Atom (nitrogen).

<u>Step 6:</u> if the central atom does not have an octet configuration, form double bonds, if necessary form triple bonds.

So, the Lewis structure of NH_3 is





<u>Step 7:</u> Check that you have the lowest FORMAL CHARGES possible for all the atoms, without violating the octet rule.

7. Lewis structure

Formal Charge for each atom is given by:

Formal Charge=(valence e^-) - (1/2 bonding e^-) - (lone electrons).

Example:

Draw the Lewis structure of NO_3^-.

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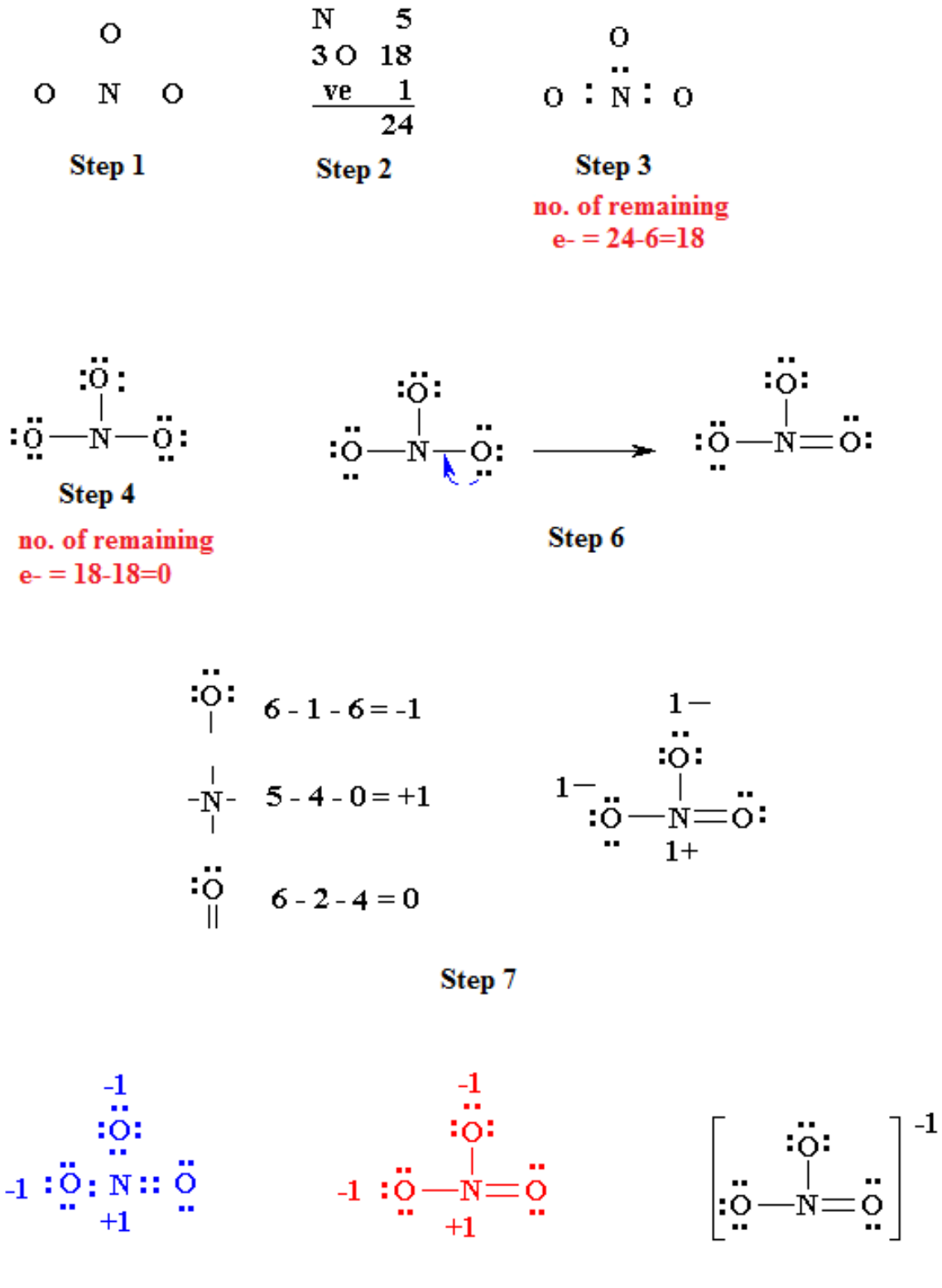
Example.

Draw the



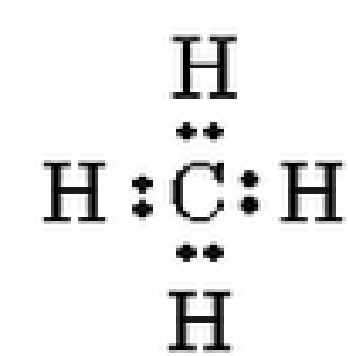
7. Lewis structure

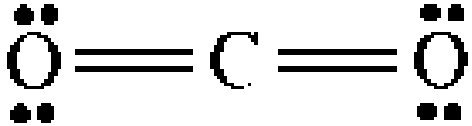
- Draw the Lewis structure of CH_4, CO_2, SO_3, NO_2^-

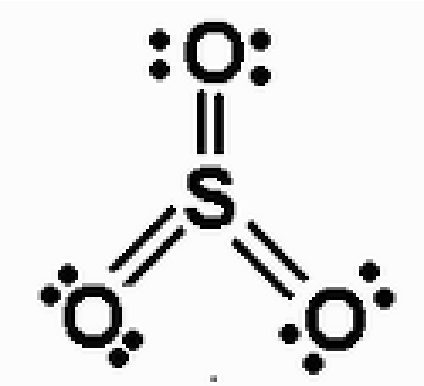
Note: if the structure you wish to draw is that of an ion, so add one additional valence electron for each negative charge or remove a valence electron for each positive charge

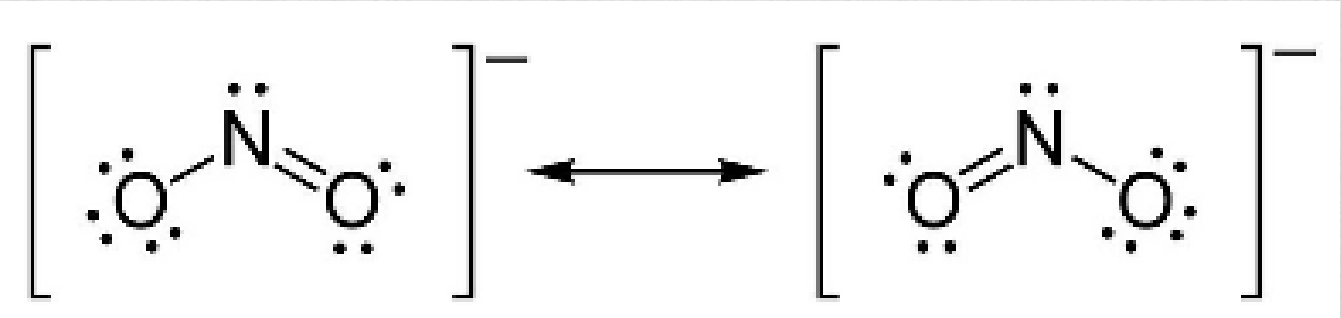
7. Lewis structure

- Draw the Lewis structure of CH_4, CO_2, SO_3.









End of Lecture 7